

Bài thực hành Lab 8.1. Bản cập nhật tháng 11 năm 2024

Mã khởi đầu: <https://github.com/google-developer-training/android-basics-kotlin-unsramble-app/tree/starter>

Bước 1. Tạo project với tên App unsramble

Cập nhật API 35

Trong file app/build.gradle.kts

```
android {  
    namespace = "com.example.unsramble"  
    compileSdk = 35  
  
    defaultConfig {  
        applicationId = "com.example.unsramble"  
        minSdk = 26  
        targetSdk = 35  
    }  
}
```

Ấn Sync để đồng bộ thư viện.

Bước 2. Cài đặt thư viện SafeArgs và Fragment

Trong file build.gradle.kts của Project

// Top-level build file where you can add configuration options common to all sub-projects/modules.

```
plugins {  
    alias(libs.plugins.android.application) apply false  
    alias(libs.plugins.kotlin.android) apply false  
    id("androidx.navigation.safeargs") version "2.8.3" apply false  
}
```

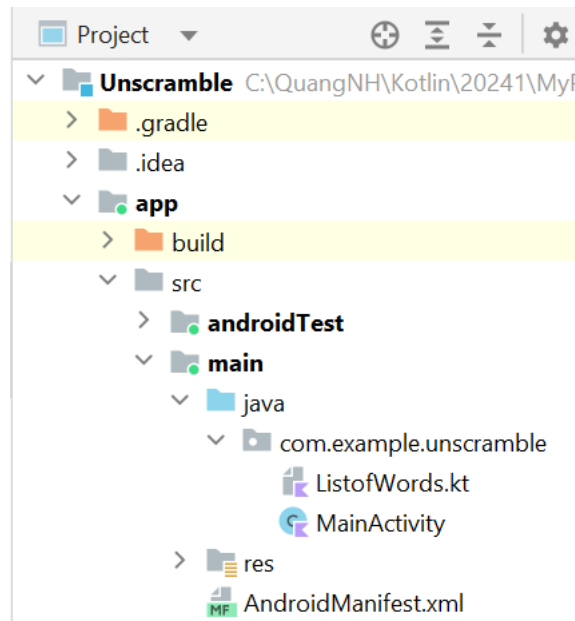
Trong file app/build.gradle.kts: bổ sung các phần bôi vàng

```
plugins {  
    alias(libs.plugins.android.application)  
    alias(libs.plugins.kotlin.android)  
    id("androidx.navigation.safeargs.kotlin")  
}  
  
android {  
    buildFeatures {  
        dataBinding = true  
    }  
}
```

```
dependencies {
    implementation("androidx.navigation:navigation-fragment-ktx:2.8.3")
    implementation("androidx.navigation:navigation-ui-ktx:2.8.3")
}
```

Bước 3. Copy file ListofWords.kt vào project

<https://github.com/google-developer-training/android-basics-kotlin-unsramble-app/blob/starter/app/src/main/java/com/example/android/unsramble/ui/game/ListofWords.kt>



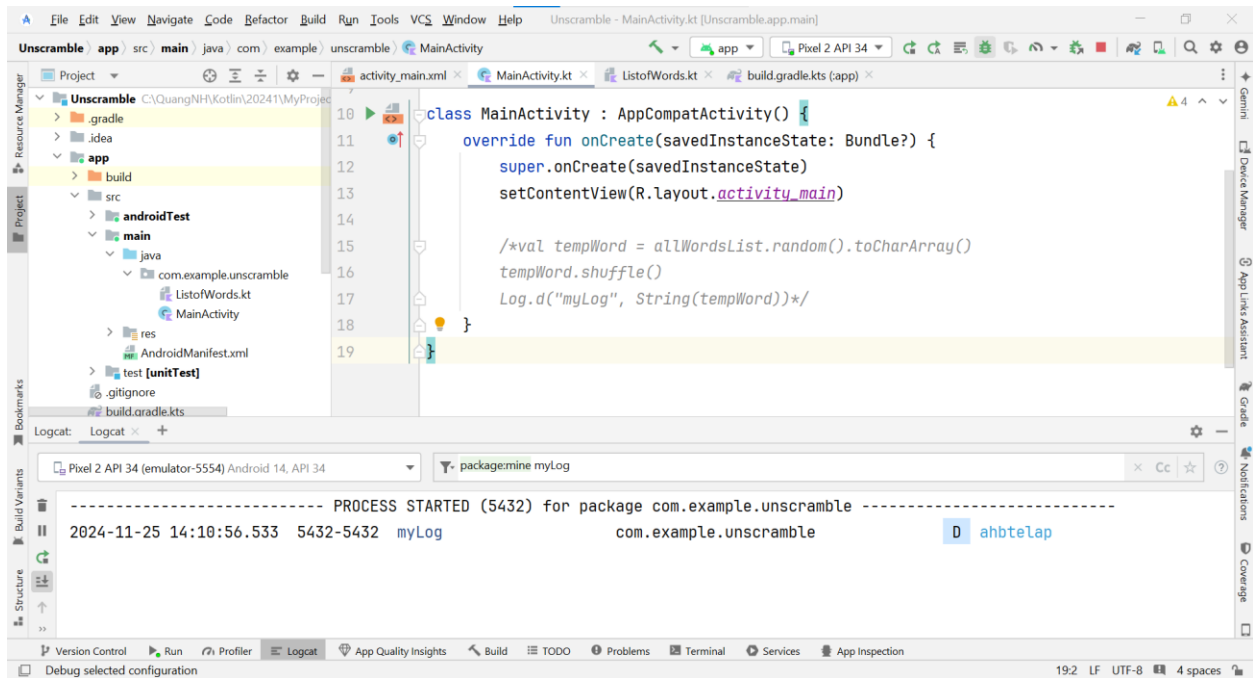
Cập nhật lại tên gói:

```
package com.example.unscramble
```

Thử nghiệm đối tượng *allWordsList* khai báo trong file *ListofWords.kt*

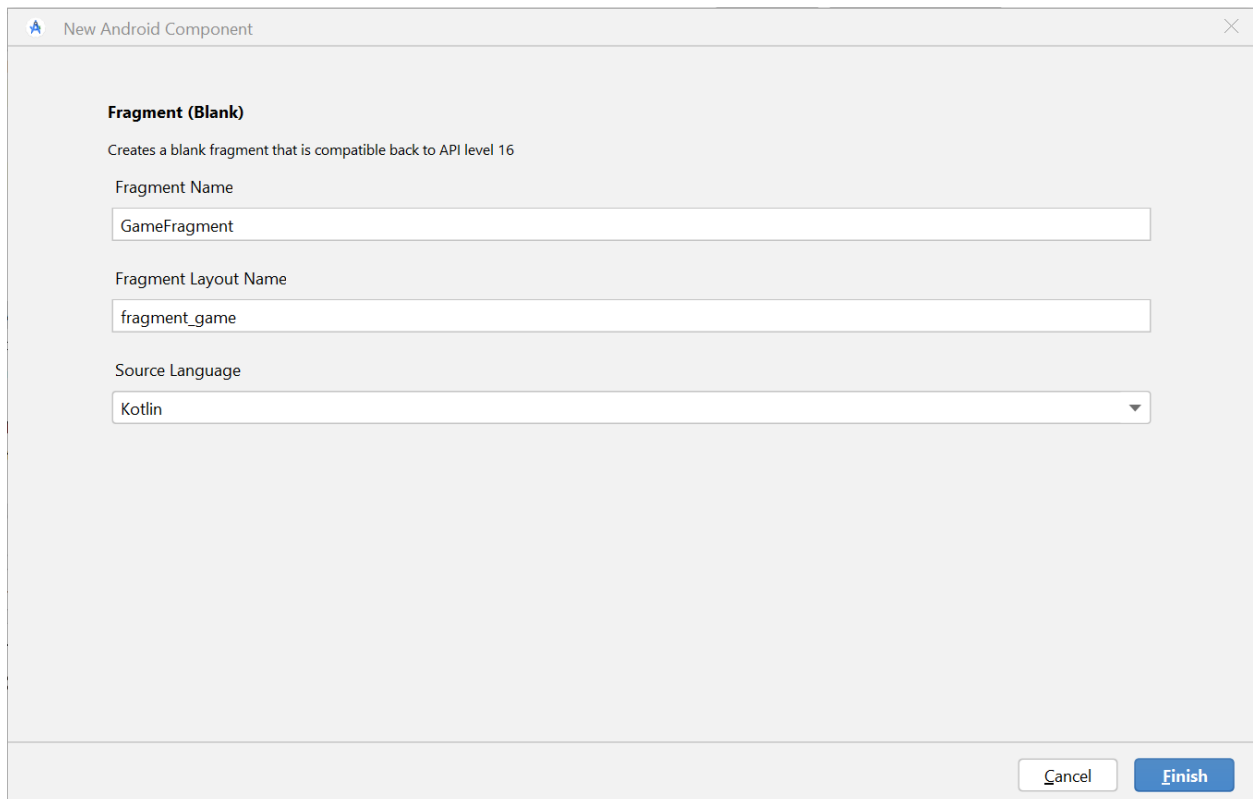
```
class MainActivity : AppCompatActivity() {
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)

        /*val tempWord = allWordsList.random().toCharArray()
        tempWord.shuffle()
        Log.d("myLog", String(tempWord))*/
    }
}
```



Bước 4. Bổ sung GameFragment

Chọn File => New => Fragment => Fragment (Blank)



```

package com.example.unscramble

import android.os.Bundle
import androidx.fragment.app.Fragment
import android.view.LayoutInflater
import android.view.View
import android.view.ViewGroup

class GameFragment : Fragment() {
    override fun onCreateView(
        inflater: LayoutInflater, container: ViewGroup?,
        savedInstanceState: Bundle?
    ): View? {
        // Inflate the layout for this fragment
        return inflater.inflate(R.layout.fragment_game, container, false)
    }
}

```

Bước 5. Thiết lập cơ chế View Binding cho GameFragment

```

package com.example.unscramble

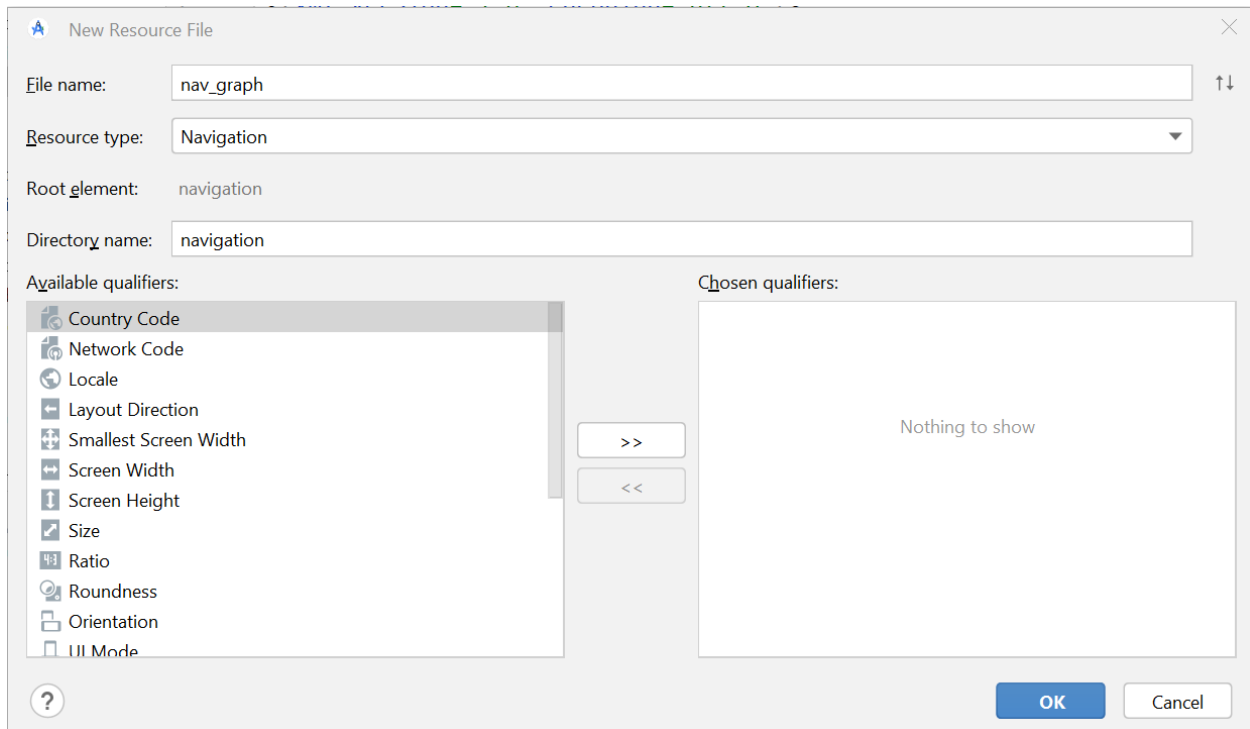
import android.os.Bundle
import androidx.fragment.app.Fragment
import android.view.LayoutInflater
import android.view.View
import android.view.ViewGroup
import com.example.unscramble.databinding.FragmentGameBinding

class GameFragment : Fragment() {
    private lateinit var binding: FragmentGameBinding
    override fun onCreateView(
        inflater: LayoutInflater, container: ViewGroup?,
        savedInstanceState: Bundle?
    ): View? {
        // Inflate the layout for this fragment
        //return inflater.inflate(R.layout.fragment_game,
        container, false)
        binding = FragmentGameBinding.inflate(inflater,
        container, false)
        return binding.root
    }
}

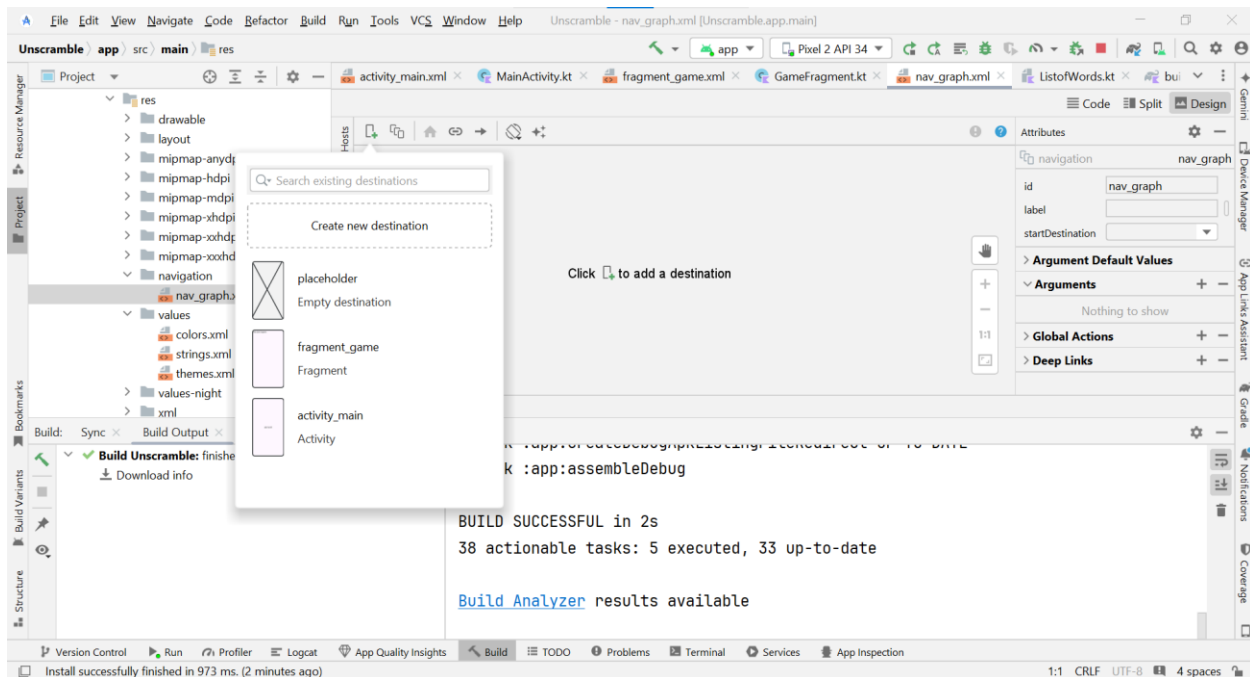
```

Bước 6. Tạo file Navigation Graph

Ấn chuột phải vào thư mục res, chọn New => Android Resource File



Bước 7. Thêm GameFragment vào nav_graph.xml



Bước 8. Tích hợp Navigation Graph vào MainActivity

Sửa lại file activity_main.xml như sau:

```
<?xml version="1.0" encoding="utf-8"?>
<FrameLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <fragment
        android:id="@+id/nav_host"

        android:name="androidx.navigation.fragment.NavHostFragment"
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        app:defaultNavHost="true"
        app:navGraph="@navigation/nav_graph" />
</FrameLayout>
```

Chạy chương trình:



Bước 9. Tạo file `dimens.xml` trong thư mục `values`

```
<?xml version="1.0" encoding="utf-8"?>
<resources>
    <dimen name="default_padding">16dp</dimen>
    <dimen name="default_margin">32dp</dimen>
</resources>
```

Bước 10. Bổ sung các chuỗi ký tự trong file `strings.xml`

```
<resources>
    <string name="app_name">Unscramble</string>
    <!-- TODO: Remove or change this placeholder text -->
    <string name="hello_blank_fragment">Hello blank
fragment</string>
    <string name="instructions">Unscramble the word using all
the letters.</string>
    <string name="skip">Skip</string>
    <string name="submit">Submit</string>
    <string name="score">Score: %d</string>
    <string name="word_count">%d of %d words</string>
    <string name="enter_your_word">Enter your word</string>
    <string name="congratulations">Congratulations!</string>
    <string name="you_scored">"You scored: %d"</string>
    <string name="exit">Exit</string>
    <string name="play_again">Play Again</string>
    <string name="try_again">Try again!</string>
</resources>
```

Bước 11. Bổ sung file `styles.xml`

```
<?xml version="1.0" encoding="utf-8"?>
<resources>
    <style name="Widget.Unscramble.TextInputLayout.OutlinedBox"
parent="Widget.MaterialComponents.TextInputLayout.OutlinedBox">
        <item name="boxStrokeErrorColor">@color/red_700</item>
        <item name="errorIconTint">@color/red_700</item>
        <item name="errorTextColor">@color/red_700</item>
        <item name="errorIconDrawable">@drawable/ic_error</item>
        <item
name="helperTextTextAppearance">@style/TextAppearance.MaterialCo
mponents.Subtitle1</item>
    </style>
</resources>
```

Bước 12. Bổ sung file colors.xml

```
<?xml version="1.0" encoding="utf-8"?>
<resources>
    <color name="black">#FF000000</color>
    <color name="white">#FFFFFFFF</color>
    <color name="indigo_200">#FF9FA8DA</color>
    <color name="indigo_500">#FF3F51B5</color>
    <color name="indigo_800">#FF283593</color>
    <color name="light_blue_200">#FF81D4FA</color>
    <color name="light_blue_700">#FF0288D1</color>
    <color name="red_700">#FFD32F2F</color>
    <color name="red_400">#FFE53535</color>
</resources>
```

Bước 13. Tạo file ic_error.xml trong thư mục drawable

```
<vector android:height="24dp" android:tint="#C62828"
    android:viewportHeight="24" android:viewportWidth="24"
    android:width="24dp"
xmlns:android="http://schemas.android.com/apk/res/android">
    <path android:fillColor="@android:color/white"
android:pathData="M11,15h2v2h-2zM11,7h2v6h-2zM11.99,2C6.47,2
2,6.48 2,12s4.47,10 9.99,10C17.52,22 22,17.52 22,12S17.52,2
11.99,2zM12,20c-4.42,0 -8,-3.58 -8,-8s3.58,-8 8,-8 8,3.58 8,8 -
3.58,8 -8,8z"/>
</vector>
```

Bước 14. Tạo giao diện cho GameFragment

```
<?xml version="1.0" encoding="utf-8"?>

<ScrollView
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <androidx.constraintlayout.widget.ConstraintLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:padding="@dimen/default_padding"
        tools:context=".ui.game.GameFragment">
```

```

<Button
    android:id="@+id/skip"
    style="?attr/materialButtonOutlinedStyle"
    android:layout_width="0dp"
    android:layout_height="wrap_content"
    android:layout_marginStart="@dimen/default_padding"
    android:layout_marginEnd="@dimen/default_padding"
    android:text="@string/skip"

app:layout_constraintBaseline_toBaselineOf="@+id/submit"
    app:layout_constraintEnd_toStartOf="@+id/submit"
    app:layout_constraintStart_toStartOf="parent" />

<Button
    android:id="@+id/submit"
    android:layout_width="0dp"
    android:layout_height="wrap_content"
    android:layout_marginTop="@dimen/default_margin"
    android:text="@string/submit"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintStart_toEndOf="@+id/skip"
    app:layout_constraintTop_toBottomOf="@+id/textField"
/>

<TextView
    android:id="@+id/textView_instructions"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="@string/instructions"
    android:textSize="17sp"
    app:layout_constraintBottom_toTopOf="@+id/textField"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintStart_toStartOf="parent"

app:layout_constraintTop_toBottomOf="@+id/textView_unscrambled_w
ord" />

<TextView
    android:id="@+id/textView_unscrambled_word"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginTop="@dimen/default_margin"
    android:layout_marginBottom="@dimen/default_margin"

    android:textAppearance="@style/TextAppearance.MaterialComponents
.Headline3"

```

```

app:layout_constraintBottom_toTopOf="@+id/textView_instructions"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintStart_toStartOf="parent"

app:layout_constraintTop_toBottomOf="@+id/word_count"
    tools:text="Scramble word" />

<TextView
    android:id="@+id/word_count"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="@string/word_count"

    android:textAppearance="@style/TextAppearance.MaterialComponents
.Headline6"

app:layout_constraintBottom_toTopOf="@+id/textView_unscrambled_w
ord"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    tools:text="3 of 10 words" />

<TextView
    android:id="@+id/score"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="@string/score"
    android:textAllCaps="true"

    android:textAppearance="@style/TextAppearance.MaterialComponents
.Headline6"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    tools:text="Score: 20" />

<com.google.android.material.textfield.TextInputLayout
    android:id="@+id/textField"

    style="@style/Widget.Unscramble.TextInputLayout.OutlinedBox"
    android:layout_width="0dp"
    android:layout_height="wrap_content"
    android:layout_marginTop="@dimen/default_margin"
    android:hint="@string/enter_your_word"
    app:errorIconDrawable="@drawable/ic_error"

    app:helperTextTextAppearance="@style/TextAppearance.MaterialComp
onents.Subtitle1"

```

```

        app:layout_constraintBottom_toTopOf="@+id/submit"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"

app:layout_constraintTop_toBottomOf="@+id/textView_instructions"
>

<com.google.android.material.textfield.TextInputEditText
    android:id="@+id/text_input_edit_text"
    android:layout_width="match_parent"
    android:layout_height="match_parent"

    android:inputType="textPersonName|textNoSuggestions"
    android:maxLength="1" />
</com.google.android.material.textfield.TextInputLayout>

</androidx.constraintlayout.widget.ConstraintLayout>
</ScrollView>
https://github.com/google-developer-training/android-basics-kotlin-unscramble-app/blob/starter/app/src/main/res/layout/game\_fragment.xml

```

Bước 15. Khai báo các biến trạng thái trong GameFragment

```

private var score = 0
private var currentWordCount = 0
private var currentScrambledWord = "test"

```

Bước 16. Bổ sung trong GameFragment các hàm sau

```

/*
 * Gets a random word for the list of words and shuffles the
 * letters in it.
 */
private fun getNextScrambledWord(): String {
    val tempWord = allWordsList.random().toCharArray()
    tempWord.shuffle()
    return String(tempWord)
}

```

Hàm này để tạo một từ mới với các ký tự được trộn từ các ký tự lấy của một từ trong từ điển.

```

/*
 * Checks the user's word, and updates the score accordingly.
 * Displays the next scrambled word.
 */
private fun onSubmitWord() {

```

```

        currentScrambledWord = getNextScrambledWord()
        currentWordCount++
        score += SCORE_INCREASE
        binding.wordCount.text = getString(R.string.word_count,
currentWordCount, MAX_NO_OF_WORDS)
        binding.score.text = getString(R.string.score, score)
        setErrorTextField(false)
        updateNextWordOnScreen()
    }

    /*
     * Skips the current word without changing the score.
     * Increases the word count.
     */
    private fun onSkipWord() {
        currentScrambledWord = getNextScrambledWord()
        currentWordCount++
        binding.wordCount.text = getString(R.string.word_count,
currentWordCount, MAX_NO_OF_WORDS)
        setErrorTextField(false)
        updateNextWordOnScreen()
    }

    /*
     * Re-initializes the data in the ViewModel and updates the
     views with the new data, to
     * restart the game.
     */
    private fun restartGame() {
        setErrorTextField(false)
        updateNextWordOnScreen()
    }

    /*
     * Exits the game.
     */
    private fun exitGame() {
        activity?.finish()
    }

    /*
     * Sets and resets the text field error status.
     */
    private fun setErrorTextField(error: Boolean) {
        if (error) {
            binding.textField.isErrorEnabled = true
            binding.textField.error = getString(R.string.try_again)
        }
    }

```

```

    } else {
        binding.textField.isErrorEnabled = false
        binding.textInputEditText.text = null
    }
}

/*
 * Displays the next scrambled word on screen.
 */
private fun updateNextWordOnScreen() {
    binding.textViewUnscrambledWord.text = currentScrambledWord
}

```

Bước 18. Bổ sung trong GameFragment hàm onCreateView

```

override fun onCreateView(view: View, savedInstanceState: Bundle?) {
    super.onCreateView(view, savedInstanceState)

    // Setup a click listener for the Submit and Skip buttons.
    binding.submit.setOnClickListener { onSubmitWord() }
    binding.skip.setOnClickListener { onSkipWord() }
    // Update the UI
    updateNextWordOnScreen()
    binding.score.text = getString(R.string.score, 0)
    binding.wordCount.text = getString(
        R.string.word_count, 0, MAX_NO_OF_WORDS)
}

```

Các vấn đề với Starter code

As you played the game, you may have observed the following bugs:

1. On clicking the Submit button, the app does not check the player's word. The player always scores points.
2. There is no way to end the game. The app lets you play beyond 10 words.
3. The game screen shows a scrambled word, player's score, and word count. Change the screen orientation by rotating the device or emulator. Notice that the current word, score, and word count are lost and the game restarts from the beginning.

Các vấn đề này sẽ được giải quyết trong bài thực hành Lab 8.1. Store data in ViewModel.